CSCI E-102 Econometrics and Causal Inference with R

Harvard Extension School

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Lecture 3

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Bias and Mean Squared Error (MSE)

Def.

Let $\hat{\theta}$ be an estimator of parameter θ of some distribution. Then

- *Bias* of the estimator is defined as

$$\text{Bias} = \mathrm{E}[\hat{\theta} - \theta].$$

- *Mean Squared Error* (MSE) of the estimator is defined as

$$\text{MSE} = \mathrm{E}\left[\left(\hat{\theta} - \theta\right)^2\right].$$

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Thm.

Let $\hat{\theta}$ be an estimator of θ. Then

$$\text{MSE} = \text{Bias}^2 + \mathrm{Var}(\hat{\theta}).$$

Consistency of Parameter Estimators

Def.

Let $\hat{\theta}_n$ be an estimator of a parameter θ of some distribution based on a sample of i.i.d. $X_1, X_2, \ldots, X_n$ from this distribution.

$\hat{\theta}_n$ is called *consistent* (or *asymptotically consistent*) estimator of θ if for any $\varepsilon > 0$,

$$\lim_{n\to\infty} P(|\hat{\theta}_n - \theta| > \varepsilon) = 0.$$

MSE and Efficiency

Recall:

If true value of parameter is θ, then the *Mean Squared Error* (MSE) of estimator $\hat{\theta}$ is defined as

$$\begin{aligned}\text{MSE}(\hat{\theta}) &\doteq \mathrm{E}\left[(\hat{\theta}-\theta)^2\right] \\ &= \underbrace{\left(\mathrm{E}[\hat{\theta}]-\theta\right)^2}_{\text{Bias}^2} + \text{Var}(\hat{\theta})\end{aligned}$$

MSE and Efficiency

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Def.
Estimator $\tilde{\theta}$ is said to be more *efficient* than $\hat{\theta}$ if

$$\mathsf{Var}(\tilde{\theta}) < \mathsf{Var}(\hat{\theta}).$$

The ratio $\mathsf{eff}(\tilde{\theta},\hat{\theta}) \doteq \mathsf{Var}(\hat{\theta})/\mathsf{Var}(\tilde{\theta})$ is called *efficiency* of $\tilde{\theta}$ relative to $\hat{\theta}$.

Note: $\tilde{\theta}$ is more efficient means $\mathsf{eff}(\tilde{\theta},\hat{\theta}) > 1$.

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Confidence Interval: Example

Example:

Consider the following random sample from a normal distribution $N(\mu, \sigma^2)$:

$$14.2, 12.1, 15.9, 17.6, 17.4, 18.2 \quad \text{i.e.} \quad \bar{X} = 15.9 \text{ and } S = 2.3563$$

Suppose we want to estimate μ, then $\mu \approx \bar{X} = 15.9$.

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Suppose we want to estimate μ, then $\mu \approx \bar{X} = 15.9$.

Assume we also want to construct 95% *confidence interval* for μ, that is, we want

$$P\left(\bar{X} - \text{ME} < \mu \leq \bar{X} + \text{ME}\right) = 0.95,$$

then

$$P\left(-\frac{\text{ME}}{S/\sqrt{n}} < \frac{\bar{X} - \mu}{S/\sqrt{n}} \leq \frac{\text{ME}}{S/\sqrt{n}}\right) = 0.95, \text{ i.e. } \text{ME} = t_{0.025,n-1}\,\frac{S}{\sqrt{n}}.$$

Therefore,

$$\mu = \bar{X} \pm t_{0.025,n-1}\,\frac{S}{\sqrt{n}} = 15.9 \pm 2.5 \text{ with } 95\% \text{ confidence.}$$

Here, $t_{0.025,n-1}$ is defined as $P(T > t_{0.025,n-1}) = 0.025$, where $T \sim t_{n-1}$.

Confidence Intervals for Mean

Def.

Given i.i.d. sample $X_1, X_2, \ldots, X_n$ from a distribution with expectation μ, $(1-\alpha)100\%$ *confidence interval* for μ is defined as

$$\mu = \bar{X} \pm \text{ME},$$

where $P\left(\bar{X} - \text{ME} < \mu \leq \bar{X} + \text{ME}\right) = 1 - \alpha$.

Claim:

$(1-\alpha)100\%$ *confidence interval* for μ is given by:

- $\mu = \bar{X} \pm z_{\alpha/2}\,\frac{\sigma}{\sqrt{n}}$, if σ is known and data comes from normal
- $\mu = \bar{X} \pm t_{\alpha/2,n-1}\,\frac{S}{\sqrt{n}}$, if data comes from normal
- $\mu = \bar{X} \pm z_{\alpha/2}\,\frac{S}{\sqrt{n}}$, if n is large

Here,

$z_{\alpha/2}$ is defined as $P(Z > z_{\alpha/2}) = \alpha/2$, where $Z \sim N(0, 1^2)$;

$t_{\alpha/2,n-1}$ is defined as $P(T > t_{\alpha/2,n-1}) = \alpha/2$, where $T \sim t_{n-1}$.

Confidence Intervals for Difference between Two Means

Def.

Given i.i.d. samples $X_1^{(1)}, X_2^{(1)}, \ldots, X_n^{(1)}$ and $X_1^{(2)}, X_2^{(2)}, \ldots, X_n^{(2)}$ from two distributions with expectations μ_1 and μ_2, respectively, $(1-\alpha)100\%$ *confidence interval* for $\mu_1 - \mu_2$ is defined as

$$\mu - \mu_2 = \bar{X}_1 - \bar{X}_2 \pm \text{ME},$$

where $P\left(\bar{X}_1 - \bar{X}_2 - \text{ME} < \mu_1 - \mu_2 \leq \bar{X}_1 - \bar{X}_2 + \text{ME}\right) = 1 - \alpha.$

Claim:

$(1-\alpha)100\%$ *confidence interval* for $\mu_1 - \mu_2$ is given by:

- $\mu_1 - \mu_2 = \bar{X}_1 - \bar{X}_2 \pm z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$, if σ_1 and σ_2 are known and data comes from normal distributions
- $\mu_1 - \mu_2 = \bar{X}_1 - \bar{X}_2 \pm t_{\alpha/2,\text{d.f.},}\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$, if data comes from normal distributions, where d.f. $\approx \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{1}{n_1-1}\left(\frac{S_1^2}{n_1}\right)^2 + \frac{1}{n_2-1}\left(\frac{S_2^2}{n_2}\right)^2}$
- $\mu_1 - \mu_2 = \bar{X}_1 - \bar{X}_2 \pm z_{\alpha/2}\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$, if n_1 and n_2 are large

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Pooled Variance: Difference between Two Means

Assume i.i.d. samples $X_1^{(1)}, X_2^{(1)}, \dots, X_n^{(1)}$ and $X_1^{(2)}, X_2^{(2)}, \dots, X_n^{(2)}$ come from two normal distributions with same variance σ^2, then

$$\begin{aligned}\sigma^2 \approx S_p^2 &\doteq \frac{1}{n_1+n_2-2}\left[\sum_{i=1}^{n_1}(X_i^{(1)}-\bar{X}_1)^2+\sum_{i=1}^{n_2}(X_i^{(2)}-\bar{X}_2)^2\right]\\ &=\frac{(n_1-1)S_1^2+(n_2-1)S_2^2}{n_1+n_2-2}.\end{aligned}$$

Then

- test statistic for $H_0 : \mu_1 - \mu_2 = \Delta_0$ vs. $H_1 : \mu_1 - \mu_2 \neq \Delta_0$ is given by:

$$\text{TS} = \frac{\bar{X}_1 - \bar{X}_2 - \Delta_0}{S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}}, \ \text{t-test}, \ \ d.f. = n_1 + n_2 - 2$$

- $(1-\alpha)100\%$ *confidence interval* for $\mu_1 - \mu_2$ is given by:

$$\mu_1 - \mu_2 = \bar{X}_1 - \bar{X}_2 \pm t_{\alpha/2, n_1+n_2-2}\, S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}$$

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Proportions: Hypotheses Testing and Confidence Intervals

Assume $X_1, X_2, \ldots, X_n$ is an i.i.d. sample from binary distribution consisting of 0's and 1's, then

- $\mu = p$ is proportion
- $\bar{X} = \hat{p}$ is sample proportion
- Var$(\hat{p}) = \frac{p(1-p)}{n}$

Note that by CLT, $\hat{p}$ tends to be approximately Normal, because $\hat{p}$ is the sample mean.

Therefore, if n is large, one can use z-test and $z_{\alpha/2}$ critical value, otherwise - Exact Binomial Test.

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Sample Covariance and Sample Correlation

Def.

Assuming i.i.d. observations $(X_1, Y_1), (X_2, Y_2) \ldots, (X_n, Y_n)$ from a joint distribution, then

- *sample variances* of X and Y:

$$s_X^2 \doteq \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2 \quad \text{and} \quad s_Y^2 \doteq \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2$$

- *sample covariance*:

$$s_{XY} \doteq \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})$$

- *sample correlation coefficient* (or *sample correlation*):

$$r_{XY} = \frac{s_{XY}}{s_X s_Y}$$

Sample Covariance and Sample Correlation

Claim:

Assuming i.i.d. observations $(X_1, Y_1), (X_2, Y_2) \dots, (X_n, Y_n)$ from a joint distribution,

$$s_{XY} \xrightarrow{P} \text{Cov}(X, Y).$$

Claim:

Assuming i.i.d. observations $(X_1, Y_1), (X_2, Y_2) \dots, (X_n, Y_n)$ from a joint distribution,

$$r_{XY} \xrightarrow{P} \text{corr}(X, Y).$$

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Simple Linear Regression: Normal Homoskedastic Error

Assumptions:
For $i \in \{1, 2, \ldots, n\}$, let

- x_i be fixed values with $\min\{x_i\} < \max\{x_i\}$
- $\varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$, where σ^2 is independent of i (*homoskedasticity*)
- $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$

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- $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$

Note:

Given x_i,

$$y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2) \text{ are independent, where } i \in \{1, 2, \ldots, n\}.$$

Then

- One can use, for example, MLE to estimate β_0 and β_1.
- Given *predictor* x_i, the expectation of *response* y_i is

$$\mathrm{E}[y_i] = \beta_0 + \beta_1 x_i.$$

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Simple Linear Regression: Least Squares

Claim:
Given a Simple Linear Regression model,

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2), \; i \in \{1, 2, \ldots, n\},$$

the Maximum Likelihood Estimators (MLE's) of β_0 and β_1 can be equivalently obtained by minimizing the sum of squared residuals,

$$S(\beta_0, \beta_1) = \sum_{i=1}^{n} (y_i - \underbrace{[\beta_0 + \beta_1 x_i]}_{\mathrm{E}[y_i]})^2,$$

and are

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} \quad \text{and } \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Note:
These estimators are often referred as *ordinary least squares* (OLS) estimators.